

## **SECTION A – CASE QUESTIONS** (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

### **CASE**

The Flying Group ("the Group") has been engaged in the toy trading business for more than 30 years. The Group has been very successful in the original equipment manufacturer ("OEM") business in which the Group receives toy manufacturing orders from overseas customers and helps them find manufacturing plants in mainland China to manufacture toys and then arranges shipments of finished products to customers.

The Group has a premium toy division that has been investing in designing, manufacturing, and building their own brand. One of the premium toys that is seeing an increase in demand is called CAR, which is an electronic remote control toy car with GPS functions. CAR is manufactured by the CAR division under the premium toy division. The Group has invested in CAR for ten years, during which continuous improvements have been made, earning the Group a few patents as well as a few best design awards in the toy industry.

CAR was designed in Hong Kong, manufactured in mainland China, and sold in both the Hong Kong and mainland China markets as a premium toy. Due to significant changes in technologies in recent years, the Group may need to invest an additional HK\$50 million to revamp CAR's remote control and GPS system. With the deteriorating economic situation, the division has had to formulate a new business strategy to improve the profitability and future growth of the business before they can have sufficient funding to revamp CAR.

The Group has been offering continuous financial support to the premium toys division because the sole shareholder of the Group was keen to build the branding of CAR within the premium toy industry. Significant investment in design, research, and development had been made to build CAR, and as a result the CAR division has incurred accumulated losses and owes HK\$40 million to the Group on an interest-free basis. The sole shareholder had hoped to improve the business of CAR to make it profitable via expansion into overseas markets; nevertheless, due to the poor economic situation and the pandemic, CAR's business has been slow to improve. Now that the sole shareholder is near retiring age, he does not want to operate the business and has also decided not to invest further in the CAR division. The Group needs to conduct a financial restructuring of the division in order to find prospective investors to further grow and develop the branding and business of CAR.

The sole shareholder of the Group has invited his friend, Peter, who is a financial consultant and CPA, to handle the financial restructuring of the CAR division. The sole shareholder has told Peter to maximise the value of the division so that the Group can recover the investments, time, and effort spent over the past ten years. The sole shareholder has promised to pay 10% of the sale consideration to Peter.

After discussing the financial situation with the general manager of the CAR division, Peter has found that the division has a significant amount of inventory in Hong Kong and mainland China that has not been formally counted by the auditors in previous years' audits, and the accounts receivable aging situation has deteriorated over the previous few years. Moreover, the CAR division purchases electronic components from an associate company of the Group at a lower cost than the market. The general manager has said that these situations are normal in business and should not affect the valuation of the CAR division.

The CAR division has its own separate set of financial statements for performance evaluation purposes. The Group has no experience in valuation because they have not conducted a financial restructuring exercise before. The financial controller of the Group has found the following financial information which may help with the valuation of the CAR division. The financial controller has also found that the beta and the expected return on the market of similar companies are 1.2 and 8%, respectively. The 10-year government bond rate is 3%. The tax rate is 16.5%. The following information summarises the key financial figures for the year ended 31 December 2021 for the CAR division:

| Extracts of financial information as at 31 December 2021 | HK\$        |
|----------------------------------------------------------|-------------|
| Sales                                                    | 100,000,000 |
| EBIT                                                     | 20,000,000  |
| Depreciation                                             | 200,000     |
| Amortisation of patent and trademark                     | 400,000     |
| Accounts receivable                                      | 20,000,000  |
| Accounts payable                                         | 10,000,000  |
| Inventory                                                | 30,000,000  |
| Patent and trademark, net                                | 4,000,000   |
| Property, plant, and equipment, net                      | 2,000,000   |
| Long-term loan from the Group                            | 40,000,000  |

In a 3-year business planning meeting, the management estimates that if the revamp of the remote control and GPS system could start in the next year, the CAR division could expect sales to increase by 30% annually. The EBIT margin would be the same as in 2021 due to an increase in the manufacturing and operating costs. The depreciation and amortisation of the patent and trademark will grow at 20%, which is in line with the annual growth of CAPEX, amounting to HK\$10 million, HK\$20 million, and HK\$20 million in each of the coming three years. The increase in accounts receivable, inventory, and accounts payable were HK\$2 million, HK\$3 million, and HK\$1 million, respectively, for the year ended 2021. The management team expects that the annual growth of accounts receivable, inventory, and accounts payable will be 20%, 30%, and 10%, respectively, compared to their respective increase for the year ended 31 December 2021. The growth rate is assumed to be 7%.

**Question 1** (10 marks – approximately 18 minutes)

Acting as the general manager of the CAR division, write a memo to the Board of Directors to formulate business growth strategies to improve profitability and future growth of CAR's branding and business. Evaluate each strategy.

Note: A maximum of 2 marks for communication skills will be awarded.

(10 marks)

**Question 2** (15 marks – approximately 27 minutes)

Acting as the financial consultant, identify and evaluate financial reconstruction approaches that are available to the Group. Recommend and justify one appropriate method to restructure the CAR division.

(15 marks)

**Question 3** (10 marks – approximately 18 minutes)

Advise Peter to identify and address ethical dilemmas about the financial reconstruction of the CAR division.

(10 marks)

**Question 4** (15 marks – approximately 27 minutes)

Evaluate the free cash flow to the firm of the CAR division in 2021 and use the most appropriate valuation method, with justification, to determine the value of the CAR division.

Note: A maximum of 2 marks for analytical skills will be awarded.

(15 marks)

\* \* \* \* \*

End of Section A

## **SECTION B – ESSAY / SHORT QUESTIONS** (Total: 50 marks)

Answer **ALL** of the following questions. Marks will be awarded for logical argumentation/ calculation and appropriate presentation of the answers.

### **Question 5** (16 marks – approximately 29 minutes)

High One Electrical Limited ("Highone") is a local distributor who distributes various overseas brands of electrical appliances such as television and washing machines in Hong Kong. Highone purchases from suppliers in Japan and Europe who bill Highone in JPY and EUR and selling to local retailers in HKD with 60 days credit. Suppliers provide 30 to 45 days credit to Highone. The cost of capital of Highone is 5%.

Below is the extract of Highone's financial statement from the previous two years.

|                      | Year  |       |
|----------------------|-------|-------|
|                      | 2021  | 2022  |
|                      | HK\$m | HK\$m |
|                      |       |       |
| Sales                | 240   | 180   |
| Cost of goods sold   | 225   | 173   |
| Purchase             | 216   | 168   |
|                      |       |       |
| Accounts receivables | 45    | 35    |
| Inventories          | 54    | 49    |
| Accounts payables    | 36    | 28    |

### **Required:**

- (a) Calculate the cash conversion cycle in days for the Year 2021 and 2022 based on the extract of Highone's financial statements. (Please use ending balance instead of average balance when calculating the turnover days of working capital). (6 marks)
- (b) In order to shorten their accounts receivables turnover days, Highone is considering implementing early settlement discounts for customers who settle their balances immediately (i.e. cash on delivery). Highone does not want the cost of such an early settlement discount to be higher than their cost of capital. Calculate the maximum early settlement discount that Highone can offer. (3 marks)

Highone is approached by retailers to convert their existing wholesale model to a consignment model. Under the consignment model, although inventories are physically stored at the retailers' locations, the title of inventories will remain with Highone and the retailers will initiate a purchase from Highone when the inventories are sold to end customers. As the retailers are no longer stocking up on any inventories they are willing to settle their purchases with Highone from 60 days to 30 days.

- (c) Without calculation, analyse the impacts of the consignment model to Highone's cash conversion cycle in days.

(2 marks)

- (d) Advise Highone the advantages, disadvantages, and considerations related to working capital management of implementing the consignment model.

(5 marks)

**Question 6 (18 marks – approximately 32 minutes)**

Mega Limited ("the Company") is a health product trading company based in Hong Kong and listed on the Main Board of the Hong Kong Stock Exchange. The Company is trying hard to reduce their greenhouse gas emissions (the majority of which is carbon dioxide ("Carbon")). The Company considers this to be a lengthy process which involves huge capital expenditure investment, as the weighted average cost of capital (WACC) is about 9% p.a.

The Company has estimated that it would take an initial investment of about HK\$8 million, including the replacement of their vehicles and renovation of their office and warehouse, in order to achieve a 30% emission reduction in the current year, although HK\$4 million of such capital expenditure is considered as necessary to support business development regardless of the emission reduction objective. The Company has adopted industry best practices by purchasing carbon credits to offset their emissions. The current price of carbon credit is HK\$350 per ton, which has been stable over the past few years and the Company does not foresee any changes in the future.

The Company is preparing the environmental section of its Environmental, Social, and Governance ("ESG") report on emissions. Below is an extract from their draft report on emissions.

| Description                       |                                                                    | Unit             | Year 2 | Year 1 |
|-----------------------------------|--------------------------------------------------------------------|------------------|--------|--------|
| Greenhouse gas (Carbon) emissions |                                                                    |                  |        |        |
| Scope 1                           | Company-owned trucks                                               | CO2 - tons       | 150    | 120    |
| Scope 2                           | Purchased electricity used in office, logistics centre, and stores | CO2 - tons       | 7,500  | 9,000  |
| Scope 3                           |                                                                    |                  |        |        |
|                                   | Business air travel                                                | CO2 - tons       | -      | 1      |
|                                   | Electricity used for fresh water processing and sewage treatment   | CO2 - tons       | 5      | 5      |
|                                   | Trucks owned by external logistic service providers                | CO2 - tons       | 90     | 50     |
| Total headcount of the Company    |                                                                    | No. of persons   | 900    | 1200   |
| Total area of the Company         |                                                                    | Square metres    | 50,000 | 50,000 |
| Total sales of the Company        |                                                                    | HK\$ in millions | 500    | 450    |

**Required:**

- (a) Calculate the carbon intensities (i.e. the total carbon emission per headcount and total carbon emission per area) for both years.  
(3 marks)
- (b) Advise the Company on the points required to be covered in the mandatory disclosures of the emission section (Environmental) of their ESG report.  
(9 marks)
- (c) The Company has been approached by a few financial institutions specialising in green financing to issue HK\$8 million green bond such that Mega could support the initial investment for the reduction of emissions. By referring to the four pillars of Green Bond Principles, advise the Company on the disclosure and reporting requirements of issuing a green bond.  
(6 marks)



**Question 7** (16 marks – approximately 29 minutes)

Rover ("the Company") is an online food ordering platform based in mainland China. The Company started the online food ordering platform ten years ago in Hong Kong, and the business has been very successful over the last few years during the pandemic. There are four major online food ordering platforms in mainland China that represent 95% of the total market share. The current market share of the Company is 20% out of 95%.

The Company is a private company with investment funded by a few venture capitalists. Thanks to the successful growth over the past few years, the latest valuation of the Company is over US\$1 billion, which has made it one of the unicorn companies. The Company would like to take advantage of its current valuation to raise funds for further expansion in Asia via acquisition and targeting the biggest players in mainland China and Hong Kong.

**Required:**

- (a) Advise the Company on the areas under the respective regulatory framework for business combination that are to be considered prior to their acquisition plan in the above jurisdictions. (8 marks)
- (b) The Company also plans to list on the Main Board in Hong Kong. In order to speed up the process, the Company would like to acquire an existing listed Company in Hong Kong with minimal operations, as the valuation is lower and the integration after acquisition will be less complicated. Advise the Company on the respective regulatory framework for such a business combination. (8 marks)

\* \* \* END OF EXAMINATION PAPER \* \* \*